UCIe Sideband Wrappers — Signal Reference

This document covers every boundary port of SB_TX_WRAPPER and SB_RX_WRAPPER. Only signals visible outside the wrapper are listed.

■ Input (green) ■ Output (orange)

1. SB_TX_WRAPPER (tx_wrapper.sv)

Parameters: FIFO_DEPTH (default 4) | SB_CLK_CYCLES_PER_MS (default 800,000)

Sub-modules: SB_FSM · ltsm_encoder · SB_PATTERN_GEN · SB_PACKET_ENCODER_MUX · SB_TX_FIFO · SB_TX_FSM_Modelling · SB_CLOCK_CONTROLLER

Signal	Width	Dir	From	To	Description
Clock / Reset					
i_clk	1	input	System / PLL	SB_TX_WRAPPER	800 MHz SB / PLL clock shared by all sub-modules
i_rst_n	1	input	System	SB_TX_WRAPPER	Active-low asynchronous reset
From LTSM — Message Request					
i_msg_valid	1	input	LTSM	SB_TX_WRAPPER	New sideband message ready to send; triggers SB_FSM IDLE→ENCODING
i_msg_id	sb_msg_id	input	LTSM	SB_TX_WRAPPER	Message identifier used to look up opcode/msgcode/has_data in SB_MSG_LUT
i_ltsm_data	64	input	LTSM	SB_TX_WRAPPER	Data payload; used only when SB_MSG_LUT[i_msg_id].has_data == 1
i_ltsm_msginfo	16	input	LTSM	SB_TX_WRAPPER	MsgInfo field embedded in header bits [55:40]
From LTSM — Control					
i_start_pattern_req	1	input	LTSM	SB_TX_WRAPPER	Enter SBINIT pattern phase; starts SB_FSM→PATTERN_GEN and arms SB_PATTERN_GEN
i_sb_ltsm_resp_rcvd	1	input	LTSM	SB_TX_WRAPPER	Genuine LTSM RESP received by RX path and relayed here; exits WAIT_RSP state
i_time_out	1	input	LTSM	SB_TX_WRAPPER	8 ms timeout from TIME_OUT_COUNTER relayed by LTSM; exits WAIT_RSP on error path
From Remote Partner — SBINIT					
i_rx_sb_pattern_samp_done	1	input	Remote partner RX side (via sideband link)	SB_TX_WRAPPER	Remote die has locked our 128 UI start-pattern (two consecutive pattern words confirmed on the remote RX FSM); stops local SB_PATTERN_GEN after 4 extra iterations
To LTSM — Status					
o_busy	1	output	SB_TX_WRAPPER	LTSM	TX path is actively processing (SB_FSM not in IDLE); LTSM must not issue a new message
o_rsp_timeout	1	output	SB_TX_WRAPPER	LTSM	Latched: 8 ms elapsed without RESP → LTSM drives state to TRAINERROR; cleared on IDLE re-entry
To TIME_OUT_COUNTER					
o_timeout_start	1	output	SB_TX_WRAPPER	TIME_OUT_COUNTER	Pulse when ltsm_encoder builds a packet → start 8-ms timer
o_timeout_msg_id	sb_msg_id	output	SB_TX_WRAPPER	TIME_OUT_COUNTER	Message ID of the packet just sent; used by counter for exclusion check
o_pattern_time_out	1	output	SB_TX_WRAPPER	TIME_OUT_COUNTER	SB_PATTERN_GEN 8 ms TX-side expiry; counter passes it through to o_time_out → LTSM
Physical Output					

Signal	Width	Dir	From	To	Description
TXCKSB	1	output	SB_TX_WRAPPER	Sideband I/O pin (remote die)	Gated 800 MHz sideband clock; active for 64 UI per packet, gated during 32-UI dead-time

2. SB_RX_WRAPPER (rx_wrapper.sv)

Parameters: FIFO_DEPTH (default 4)

Sub-modules: SB_RX_FIFO · SB_RX_FSM · ltsm_decoder · sb_error_handler

Signal	Width	Dir	From	To	Description
Clock / Reset					
i_clk	1	input	System	SB_RX_WRAPPER	800 MHz SB clock
i_rst_n	1	input	System	SB_RX_WRAPPER	Active-low asynchronous reset; decoder uses ~i_rst_n (active-high)
From SerDes (Deserialiser)					
i_de_ser_done	1	input	SerDes	SB_RX_WRAPPER	Pulse: new 64-bit word has been captured by the FIFO write port; FSM receives pulse only (no data)
i_rx_word	64	input	SerDes	SB_RX_WRAPPER	Received 64-bit word from deserialiser; same bus feeds both FIFO header and data write ports; phase controlled by FSM
To LTSM — Decoded Message					
o_msg_valid	1	output	SB_RX_WRAPPER	LTSM / sb_error_handler	Complete packet committed to RX FIFO and ready for processing; sourced from SB_RX_FSM o_msg_valid (= w_msg_valid)
o_msg_id	sb_msg_id	output	SB_RX_WRAPPER	LTSM / TIME_OUT_COUNTER	Decoded message enum value from ltsm_decoder; also assigned to o_timeout_rx_msg_id
o_msginfo	16	output	SB_RX_WRAPPER	LTSM / TIME_OUT_COUNTER	MsgInfo field [55:40] from header; also assigned to o_timeout_rx_msginfo
o_rx_data	64	output	SB_RX_WRAPPER	LTSM	Data flit forwarded by decoder; valid only when the decoded message has_data == 1
o_decoder_valid	1	output	SB_RX_WRAPPER	LTSM	ltsm_decoder o_valid: message found in SB_MSG_LUT; 1 cycle after i_pckt_valid
To LTSM — Parity Errors					
o_cp_error	1	output	SB_RX_WRAPPER	LTSM	Control Parity mismatch → fatal UIE; combinational from sb_error_handler; active same cycle as w_msg_valid
o_dp_error	1	output	SB_RX_WRAPPER	LTSM	Data Parity mismatch → fatal UIE; only meaningful when i_has_data == 1
To Remote Partner TX Side — SBINIT					
o_rx_sb_start_pattern	1	output	SB_RX_WRAPPER	LTSM (which instructs local TX side to start sending pattern to remote partner)	First pattern word detected from remote partner; LTSM uses this to trigger its own TX start-pattern phase toward the remote die
o_rx_sb_pattern_samp_done	1	output	SB_RX_WRAPPER	Sent to remote partner TX side (via LTSM / sideband control); TIME_OUT_COUNTER	Local RX FSM has locked 128 UI of the remote die's start-pattern; this signal travels back to the remote die's TX wrapper as its i_rx_sb_pattern_samp_done; also stops local 8-ms counter
To TIME_OUT_COUNTER					

Signal	Width	Dir	From	To	Description
o_timeout_stop	1	output	SB_RX_WRAPPER	TIME_OUT_COUNTER i_stop	Any valid decoded message → stop 8-ms timer; sourced from ltsm_decoder o_timeout_stop
o_timeout_rx_msg_id	sb_msg_id	output	SB_RX_WRAPPER	TIME_OUT_COUNTER i_rx_msg_id	RX msg_id forwarded for stall-condition 2 check (SB_RECAL_TRACK_TX_ADJUST_RESP)
o_timeout_rx_msginfo	16	output	SB_RX_WRAPPER	TIME_OUT_COUNTER i_rx_msginfo	RX MsgInfo forwarded for stall-condition 1 (== 16hFFFF) and condition 2 ([1:0]==2b11)
Debug / Status					
o_fifo_overflow	1	output	SB_RX_WRAPPER	Debug	RX FIFO write attempted while full; should never fire in a correctly-sized system
o_de_ser_done_sampled	1	output	SB_RX_WRAPPER	SerDes	Registered ack back to deserialiser; one-cycle delayed copy of i_de_ser_done